

Kruskal's Algorithm

Kruskal's Algorithm finds the **Minimum Spanning Tree (MST)** by first sorting all edges according to their weights. It then selects the smallest edge that **does not create a cycle**. The `find_parent()` and `union_set()` functions are used to detect and avoid cycles. The process continues until $V-1$ edges are selected.

Time Complexity: $O(E \log E)$

Space Complexity: $O(V)$

Output :-



```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\shiva\Desktop\UNI Coding\DAA> cd "c:\Users\shiva\Desktop\UNI Coding\DAA"
Edges in Minimum Spanning Tree:
0 - 1 : 2
1 - 2 : 4
1 - 4 : 5
0 - 3 : 6
Total Minimum Cost = 17
PS C:\Users\shiva\Desktop\UNI Coding\DAA>
```